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PERFORMANCE SPECIFICATION

OBJECTIVE LENS ASSEMBLY, FOR
NIGHT VISION GOGGLES, AN/PVS-5( )

This specification is approved by the U.S. Army Communications-Electronics Command (CECOM), Department of the Army, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Objective Lens Assembly, part of the Night Vision Goggles, AN/PVS-5( ) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, and deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703-5023, by using the Standardization Document Improvement Proposal (DD Form 1426) or by letter.

AMSC N/A FSC 5855

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2.1.1 Reference documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.b).

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

A3144480 - Objective Lens Assembly, F/1.05
and all associated drawings
A3144481 - Lens Cell Assembly, Objective, F/1.05
A3144482 - Collar, Mounting, Objective Lens

(Copies of specification, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The Objective Lens Assembly is a part of the Night Vision Goggles, AN/PVS-5( ).

3.1.1 First article test inspection. When specified in the contract or purchase order, the contractor shall furnish assemblies for first article test (FAT) inspection and approval (see 6.2.c).

3.2 Construction. The Objective Lens Assembly, referred to hereinafter as the objective lens assembly, shall be constructed to comply with the performance requirements in the contract, drawing A3144480 as applicable, and as specified herein. The Objective Lens Cell Assembly, which is part of the objective lens assembly and is referred to hereinafter as the cell assembly, shall be in accordance with drawing A3144481 as applicable and as specified herein. The Objective Lens Mounting Collar, which is part of the objective lens assembly and is referred to hereinafter as the mounting collar, shall be in accordance with drawing A3144482 as applicable and as specified herein.

3.3 Parts and materials. The contractor shall select parts and materials fully capable of meeting all operational and environmental requirements specified herein. The materials specified on the applicable drawings are recommended, but not mandatory. Verification that the assembly meets the overall performance requirements shall be the governing acceptance standard.

3.4 Performance characteristics.

3.4.1 Equivalent focal length. The equivalent focal length of the objective lens assembly shall be 26.8 millimeters (mm) ± 0.4.

3.4.2 Flange focal distance. The flange focal distance of the objective lens assembly shall be 0.2560 inch +.0010/-.0000 where the measurement is taken from the side of the 1.763 inch mounting collar diameter closest to the focal plane (see A3144480).

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3.4.3 Linear distortion. The linear distortion of the objective lens assembly at the edge of the 18 mm format shall be between 2.5 percent and 8.5 percent barrel distortion.

3.4.4 Field of view. The field of view of the objective lens assembly shall be 40^{o} +1/-3.

3.4.5 F/number. The F/number of the objective lens assembly shall be no greater than F/1.1.

3.4.6 T/number. The T/number of the objective lens assembly shall be no greater than T/1.25.

3.4.7 Stray light. The stray light of the objective lens assembly shall be no greater than 2.5 percent.

3.4.8 Modulation transfer function (MTF). The MTF values of the objective lens assembly shall not be less than those specified in Table I.

Table I. Modulation transfer function.

Orientation	Spatial Frequency[1]			
	10 lp/mm	20 lp/mm	30 lp/mm	40 lp/mm
on-axis	76%	62%	50%	40%
14° off-axis radial[2]	60%	12%	0%	0%
14° off-axis tangential[2]	60%	12%	0%	0%

[1]in units of line pairs per millimeter (lp/mm)
[2]unrefocused from on-axis measurement

3.4.9 Field flatness/field tilt. The field flatness/field tilt of the objective lens assembly shall be within 0.3 mm across the entire 18 mm format for radial and/or tangential focus.

3.4.10 Relative illumination. At full aperture, the image illuminance at the edge of the format shall be no less than 30 percent of the image illuminance on axis.

3.4.11 Weight. The weight of the objective lens assembly, without the protective cap (see drawing A3144484), shall not exceed 104 grams.

3.5 Environmental characteristics.

3.5.1 Temperature extremes. The objective lens assembly shall not be damaged (see 3.9.1) by storage in any temperature from $+23^{o}C$ to $+68^{o}C$ or from $+23^{o}C$ to $-51^{o}C$. The objective lens assembly shall not be damaged (see 3.9.1) after being subjected to repetitive temperature changes between $+23^{o}C$ to $+65^{o}C$ in a 5 minute period and between $+23^{o}C$ to $-51^{o}C$ in a 5 minute period. Soak time at temperature extremes shall be sufficient for the entire objective lens assembly to attain stabilization. Time for temperature stabilization is defined as the amount of time required for the most internal part of the unit under test to achieve the specified temperature.

3.5.2 Humidity. The objective lens assembly shall not be damaged (see 3.9.1) when subjected to an atmosphere with a relative humidity of 95 percent ± 5 over a temperature range of $20^{o}C$ to $65^{o}C$.

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3.5.3 Vibration. The objective lens assembly shall not be damaged (see 3.9.1) by vibrations in the frequency range of 10 hertz (Hz) to 55 Hz with a double amplitude of 0.050 inch ± 0.005.

3.5.4 Immersion. The objective lens assembly shall not be damaged (see 3.9.1) after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes. The objective lens assembly shall be pre-conditioned by exposure to a temperature of 45°C ± 3 C° for a period of 2 hours prior to being immersed in the fresh water at a temperature of 18°C ± 5 C°.

3.6 Treatment and painting. Treatment and painting shall be anti-reflective, shall be rust resistant and as specified on the applicable drawings unless otherwise specified in the contract or purchase order.

3.7 Marking. Marking shall be in accordance with the drawing unless otherwise specified in the contract or purchase order.

3.8 Workmanship. Workmanship standards shall be such that the objective lens assembly will meet all requirements of this specification and any referenced specifications or drawings.

3.9 Technical interpretations. The following technical interpretations are, when referenced in sections 3, mandatory for this specification.

3.9.1 Damage. Damage is defined as: condensation, fogging, or moisture on any internal surface; failure of any finish; dents, breakage, or failure of any hardware, connection, interface, or component.

3.9.2 Ambient (room) temperature. Ambient (room) temperature is defined as 23°C ± 10 C°.

3.9.3 g. g is defined as the magnitude of the earth’s mean surface gravitational acceleration with a value of 32.2 feet per second per second.

3.9.4 Optical cell. The optical cell is defined as the cylindrical part of the cell assembly which is in direct contact with the lens elements and spacers, holds the lens elements and spacers in their fixed relative positions, and has no external pins, sleeves, flanges, retainers, or other surfaces or fittings attached to it in any manner.

3.9.5 Environmental temperature. Environmental chamber temperature shall be controlled to within ±5C°.

4. VERIFICATION

4.1 Classification of inspections. Inspections shall be classified as follows:

a. First article test inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article test inspection. When specified in the contract or purchase order (see 6.2.c), first article test (FAT) inspection shall be performed by the contractor. The quantity of FAT inspection samples shall be as specified in the contract or purchase order (see 6.2.d) and shall be selected from the first production lot.

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4.2.1 First article testing. Each objective lens assembly in the FAT lot shall be subjected to the inspections of Table II. Samples selected from the lot shall be subjected to the inspections of Table III. Sample size shall be as specified in the contract or purchase order (see 6.2.c). Failure of any inspection shall be cause for rejection of that objective lens assembly and may be cause for failure of the FAT inspection. Inspections may be performed in any order except that the Modulation transfer function test shall be performed last.

Table II. Unit first article test inspection.

Test	Requirement Paragraph
Equivalent focal length	3.4.1
Flange focal distance	3.4.2
Linear distortion	3.4.3
Field of view	3.4.4
F/number	3.4.5
T/number	3.4.6
Stray light	3.4.7
Modulation transfer function	3.4.8
Field flatness/field tilt	3.4.9
Relative illumination	3.4.10

Table III. Sample first article test inspection.

Test	Requirement Paragraph
Temperature extremes	3.5.1
Humidity	3.5.2
Vibration	3.5.3
Immersion	3.5.4
Weight	3.4.11

4.2.2 Disposition of FAT inspection samples. After successful completion of the FAT inspection, the FAT inspection samples shall be refurbished and delivered in accordance with the contract or purchase order (see 6.2.d).

4.3 Conformance inspection. Requirements for conformance inspection shall be as specified in the contract or purchase order (see 6.2.e). Unless otherwise specified, all tests shall be performed at ambient temperature (see 3.9.2).

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2.g). When actual packaging of materials is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory control Point’s packaging activity within the Military Department of Defense Agency, or within the Military Department’s System Command. Packaging data retrieval is available from the managing Military Department Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The objective lens assembly is part of the AN/PVS-5( ) Night Vision Goggles, used for general purpose viewing during periods of darkness.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Statement specifying that FAT testing is or is not required and, if so, the sampling requirements for FAT.

d. Quantity and schedule for first article test (FAT) inspection and disposition of the FAT inspection samples.

e. Statement identifying that the contractor must submit his proposal for conformance inspection to include the specific requirements to be verified and the sampling plan to be used to provide the desired confidence level.

f. Necessary actions by the contractor in the event of a lot failure.

g. Packaging requirements (see section 5).

6.3 Definitions. See 3.9.

6.6 Subject/term/keyword listing.

Objective lens assembly
Objective lens cell assembly
Objective lens mounting collar

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

Custodian:
Army - CR
Air Force - 11

Review Activity:
Army - AT
Air Force - 99

Preparing Activity:
Army - CR

Project 5855-0195

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49157B	**2. DOCUMENT DATE (YYMMDD)** 990304

3. DOCUMENT TITLE
Objective Lens Assembly, for Night Vision Goggles, AN/PVS-5()

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9129 (2) AUTOVON 992-9129
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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